



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

UNIT TEST

11-07-2021

JEE(Main + Advance) : Nurture Course (PHASE : S(TEPS))

Test Type : UNIT TEST

Test Pattern : JEE (Advanced)

TEST DATE : 11-07-2021

ANSWER KEY

PHYSICS

SEC-1	Q.	1	2	3	4				
	A.	A	B	B	D				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	B,C	B,D	C,D	B,C,D	A,B	A,B	A,D	A,B,C
SEC-3	Q.	1	2	3	4	5	6		
	A.	2.00	300.00	1.37 to 1.38	28.28	2329	610 to 611		

CHEMISTRY

SEC-1	Q.	1	2	3	4				
	A.	C	D	D	C				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,D	B,C	C	B,C,D	A,B,C,D	A,C,D	A,B,D	A,B,C,D
SEC-3	Q.	1	2	3	4	5	6		
	A.	5.00	2.66 to 2.67	6.00	8.00	7.00	1.50		

MATHEMATICS

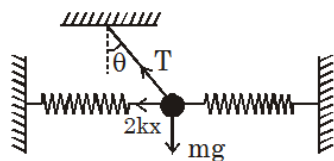
SEC-1	Q.	1	2	3	4				
	A.	B	B	D	A				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,C	A,C	A,B	B,D	A,B,C	C,D	D
SEC-3	Q.	1	2	3	4	5	6		
	A.	0.50	20.00	0.75	1.25	0.80	5.50		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A)



$$\tau = -(mgl\theta + 2kx\ell\cos\theta)$$

$$\tau \approx -(mgl\theta + 2k\ell \times \ell\theta)$$

$$\tau = m\ell^2\alpha$$

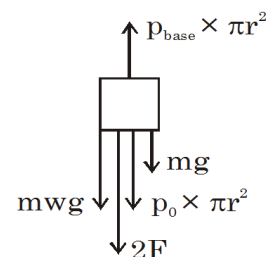
$$\alpha = -\left[\frac{g}{\ell} + \frac{2k}{m}\right]\theta$$

2. Ans (B)

Conceptual

3. Ans (B)

$$p_{\text{base}} = (p_0 + 6\rho g)$$



$$\Rightarrow 6\rho_w g \times \pi r^2 = m_w g + mg + 2F$$

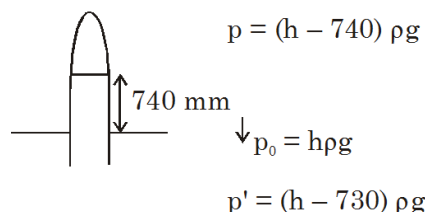
$$F = \frac{1}{2} (6.0 \times 10^4 \times \pi \times 1.5^2 - 25 \times 10^3 - \frac{2}{3} \pi \times$$

$$1.5^3 \times 10^3 \times 10 - \pi \times 5 \times 10 \times \left(\frac{2}{100}\right)^2)$$

$$\frac{10}{2} \left(6. \times 2.25\pi - 2.5 - \pi \times 2 \times 10^{-6} - \frac{2\pi}{3} \times 3.375\right)$$

$$= \frac{8.75}{2} \pi \times 10^4 \text{ N}$$

4. Ans (D)



$$pv = p'v'$$

$$(h - 740) \times 60 = (h - 730) \times 40$$

$$\frac{3h}{2} - 1110 = h - 730$$

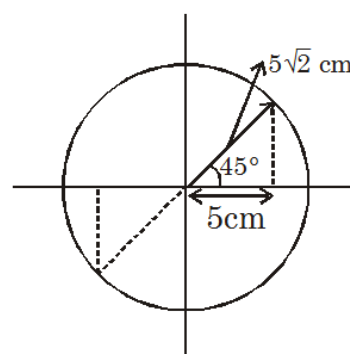
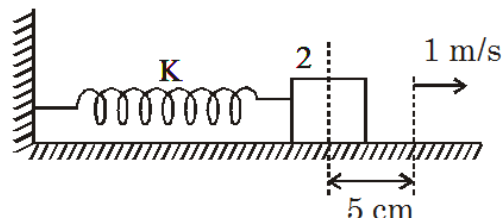
$$\frac{h}{2} = 1110 - 730 = 380$$

$$\Rightarrow h = 760 \text{ mm}$$

PHYSICS

SEC-2

1. Ans (B,C)



$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.5}} = 20$$

$$\theta = \omega t = 20 \times \frac{\pi}{20} = \pi \text{ rad} \Rightarrow S = -5 \text{ cm}$$

at x_{max}

$$\frac{\pi}{4} = \omega t = 20t ; t = \frac{\pi}{80} \text{ sec}$$

$$u = \omega \sqrt{A^2 - x^2}$$

$$1 = \sqrt{\frac{200}{0.5}} \sqrt{A^2 - \left(\frac{1}{20}\right)^2}$$

$$\frac{1}{400} = A^2 - \frac{1}{400}$$

$$A^2 = \frac{1}{200} ; A = \frac{1}{10\sqrt{2}} \text{ m} = 5\sqrt{2} \text{ cm}$$

$$u_{\text{max}} = A\omega = \frac{5\sqrt{2}}{100} \times 20 = \sqrt{2} \text{ m/s}$$

2. Ans (B,D)



$$t = 0 \Rightarrow L_i = L_f$$

$$m\ell^2 \times \theta_0 \omega_0 = m(\ell + \Delta\ell)^2 \theta \times \omega$$

$$\Rightarrow \theta_0 \ell^{3/2} = \theta(\ell + \Delta\ell)^{3/2}$$

$$\Rightarrow \theta \approx \theta_0 \left(1 - \frac{3}{2} \frac{\Delta\ell}{\ell}\right)$$

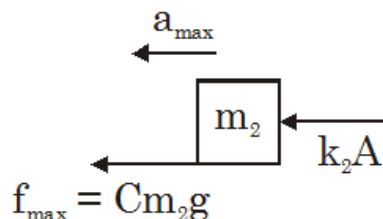
$$E = \frac{1}{2} m \times \ell^2 \times \theta_0^2 \times \frac{g}{\ell}$$

$$E = \frac{1}{2} mg\ell\theta_0^2 \Rightarrow E \downarrow$$

$$E = \frac{1}{2} mg(\ell + \Delta\ell) \times \frac{\theta_0^2 \ell^3}{(\ell + \Delta\ell)^3}$$

$$E' = \frac{1}{2} mg\ell\theta^2 < E_0$$

3. Ans (C,D)



$$a_{\max} = \left(\frac{k_1 + k_2}{m_1 + m_2} \right) A, \omega = \sqrt{\frac{k_1 + k_2}{m_1 + m_2}},$$

$$T = 2\pi \sqrt{\frac{m_1 + m_2}{k_1 + k_2}}$$

$$f_{\max} + k_2 A = m_2 \left(\frac{k_1 + k_2}{m_1 + m_2} \right) A$$

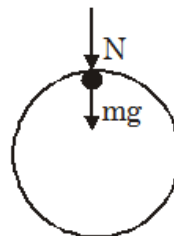
$$Cm_2 g = \left[m_2 \left(\frac{k_1 + k_2}{m_1 + m_2} \right) - K_2 \right] A$$

$$A = \frac{Cm_2 g(m_1 + m_2)}{k_1 m_2 - k_2 m_1}$$

4. Ans (B,C,D)

Momentum in vertical direction is not conserved.

Energy is conserved.



$$\text{At top } N + mg = \frac{mv_{\text{rel}}^2}{R}$$

$$-2mgR = \frac{1}{2} \mu (v_{\text{rel}}^2 - u_{\text{rel}}^2)$$

$$-2mgR = \frac{1}{2} \mu (u_{\text{rel}}^2 - u_0^2)$$

$N = mg$ because trolley is lifted

$$\Rightarrow v_{\text{rel}}^2 = 2gR$$

$$\Rightarrow v_0 = \sqrt{10gR}$$

5. Ans (A,B)

$\frac{P}{T} = \text{constant} \Rightarrow \text{Isochoric process.}$

$$Q = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 1 \times \frac{25}{3} \times (100 - 0)$$

6. Ans (A,B)

$$\frac{\frac{F}{A} + p_0}{\rho g} = \frac{p_0}{\rho g} + \frac{v^2}{2g}$$

$$F = \frac{1}{2} \rho v^2 A$$

PHYSICS

SEC-3

1. Ans (2.00)

The x coordinates of the particles are

$$x_1 = A_1 \cos \omega t, x_2 = A_2 \cos \omega t$$

$$\text{Separation} = x_1 - x_2 = (A_1 - A_2) \cos \omega t = 12 \cos \omega t$$

$$\text{Now } x_1 - x_2 = 6 = 12 \cos \omega t$$

$$\Rightarrow \omega t = \frac{\pi}{3} \Rightarrow \frac{2\pi}{12} \cdot t = \frac{\pi}{3}$$

$$\Rightarrow t = 2s$$

2. Ans (300.00)

$$a = \frac{m_1 - m_2}{m_1 + m_2} g = \frac{3 - 1}{3 + 1} g = 5 \text{ m/s}^2$$

$$v = at = 10 \text{ m/s}$$

$$P = Fv = 30 \times 10 = 300 \text{ W}$$

3. Ans (1.37 to 1.38)

$$\eta = 3 \times 10^{-5} \times 0.9 \times 10^3 = 2.7 \times 10^{-2} \text{ Pas}$$

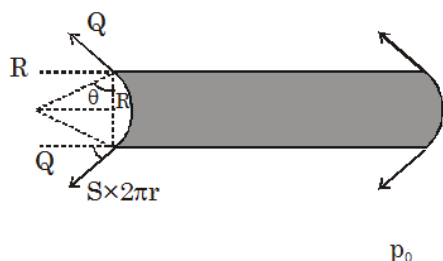
$$F = \eta A \frac{dv}{dy}$$

$$= 2.7 \times 10^{-2} \times 2\pi \times 75 \times 10^{-3} \times 300 \times 10^{-3} \times \frac{6}{0.1 \times 10^{-3}}$$

$$P = Fv = 2.7 \times \pi \times 9 \times 6 \times 3 \text{ W}$$

$$= 1373.43 \text{ kW}$$

4. Ans (28.28)



$$-S \times 2\pi r \cos\theta \times 2 = ma$$

$$\cos\theta = \frac{r}{R}$$

$$a = -\frac{4\pi S r^2}{mR}$$

$$a = -\frac{4\pi S r^2}{mR} \times 2x = \omega^2 x$$

$$(R - x)^2 + r^2 = R^2$$

$$r^2 = 2Rx$$

$$R = \frac{r^2}{2x}$$

$$\Rightarrow \omega = \sqrt{\frac{8\pi S}{m}} = \sqrt{\frac{8\pi \times 0.01}{\pi \times 10^{-6} \times 0.1 \times 10^3}}$$

$$= 20\sqrt{2} \text{ rad/s}$$

Alternate solution

$$p_0 \left(p_0 - \frac{2S}{R} \right) \left(p_0 + \frac{2S}{R} \right) p_0$$

$$\Delta p = \frac{4S}{R} = \frac{4S}{r^2} \times 2x$$

$$F_{\text{net}} = \Delta p A = ma = \rho A \ell a$$

$$a = \frac{\Delta p}{\rho \ell} = -\frac{8Sx}{\rho r^2 \ell}$$

$$\omega = \sqrt{\frac{8S}{\rho r^2 \ell}}$$

5. Ans (2329)

Let the mercury goes down by x then for gas

$$\frac{75 \times 10}{nR \times 300} = \frac{(75 - 2x)(10 + x)}{nR \times 510}$$

$$2x^2 + 95x - 525 = 0$$

$$x = 5 \text{ cm}$$

$$W = W_{\text{atm}} + W_g$$

$$= P_{\text{atm}} AX + (\Delta m)g [X]$$

$$W = (0.75 \times 13.6 \times 10^3 \times 10)A \times$$

$$\frac{5}{100} - \left(\frac{13.6 \times 10^3 \times 5}{100} \right) \times 10 \times \frac{5}{100}$$

$$W = 5440 \text{ A}$$

$$\Delta U = nC_v \Delta T$$

$$n = \frac{PV}{RT} = \frac{408}{100} \text{ A} = (17850) \text{ A}$$

6. Ans (610 to 611)

$$mg = \frac{2}{3} \pi r^3 \rho g$$

$$m = \frac{2}{3} \times \pi \times (6)^3 \times 1.35$$

$$= \pi \times 216 \times 0.9 \text{ gm}$$

$$= 610.41 \text{ gm}$$

CHEMISTRY

SEC-1

2. Ans (D)

$$\lambda = \frac{h}{\sqrt{2M_p eV}}$$

$$M_\alpha = 4M_p$$

$$\lambda_2 = \frac{h}{\sqrt{2M_\alpha 2eV}}$$

$$q_\alpha = 2e$$

$$\lambda_2 = \frac{h}{\sqrt{2M_p 8eV_2}} \text{ if } \lambda_2 = \lambda \text{ then } 8V_2 = V$$

$$V_2 = \frac{V}{8}$$

3. Ans (D)

Milli-equivalent of NH_3 = mill. eq. of

$\text{HNO}_3(\text{given}) - m_{\text{eq}} \text{ of } \text{HNO}_3(\text{remaining})$

$$\therefore m_{\text{eq. of } \text{HNO}_3(\text{remaining})} = m_{\text{eq. of } \text{Ca}(\text{OH})_2}$$

$$\Rightarrow m_{\text{eq. of } \text{HNO}_3(\text{remaining})} = M \times V_{\text{mL}} \times n_f$$

$$m_{\text{eq. of } \text{HNO}_3(\text{remaining})} = 0.5 \times 30 \times 2$$

$$\therefore m_{\text{eq. of } \text{HNO}_3(\text{remaining})} = 30$$

$$\therefore m_{\text{eq. of } \text{NH}_3} = m_{\text{eq. of } \text{HNO}_3(\text{given})} = m_{\text{eq. of } \text{HNO}_3(\text{remaining})}$$

$$\text{HNO}_3(\text{remaining})$$

$$= 200 \times 0.25 \times 1 - 30$$

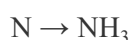
$$= 20$$

$$\therefore m_{\text{eq. of } \text{NH}_3} = 20$$

$$n_f \text{ of } \text{NH}_3 \text{ in aq. solution} \Rightarrow (n_f = 1)$$

$$\therefore \text{millimole} \times n_f = 20 \quad \{n_f = 1\}$$

$$\therefore m \text{ moles}_{(\text{NH}_3)} = 20$$



$$\therefore m \text{ moles of N} = m \text{ moles of } \text{NH}_3$$

$$\therefore \text{mmoles of N} = 20$$

$$\therefore \text{moles of N} = 20 \times 10^{-3} \times 14 \text{ g}$$

$$\therefore \text{Mass of N} = 0.280 \text{ g}$$

$$\therefore \% \text{N} = \frac{\text{Mass of N}}{\text{Mass of organic comp.}} \times 100$$

$$= \frac{0.280}{0.7} \times 100$$

$$= 40\% \quad \boxed{\text{Ans. D}}$$

CHEMISTRY

SEC-2

4. Ans (B,C,D)

90 gm glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)

$$\therefore \text{moles of glucose} = \frac{90}{180} = \frac{1}{2} \text{ moles}$$

$$\left[\frac{\text{C}_6\text{H}_{12}\text{O}_6}{180 \text{ g/mole}} \right]$$

$$\text{Total mass of solution} = (800 + 90) = 890 \text{ gm}$$

$$\text{Volume of solution} = \frac{\text{Mass of solution}}{\text{density of solution}}$$

$$= \frac{890}{\left(\frac{89}{80}\right)} = 800 \text{ mL} = 0.8 \text{ L}$$

$$\text{A) } \% \frac{w}{w} = \frac{\text{wt. of solute (glucose)}}{\text{wt. of solution}} \times 100$$

$$= \frac{90}{890} \times 100 = \frac{900}{89} \% \quad \frac{w}{w} \quad (\text{A}) \rightarrow$$

Incorrect

$$\% \frac{w}{w} = \frac{900}{89} \%$$

$$\text{B) } \% \frac{w}{v} = \frac{\text{wt. of solute (gm)}}{\text{vol. of solution (mL)}} \times 100$$

$$= \frac{90}{800} \times 100$$

$$= \frac{90}{8} \% \quad \frac{w}{v} \quad (\text{B}) \rightarrow \text{Correct}$$

$$\text{C) } m = \frac{\text{moles of solute}}{\text{mass of solvent } \text{H}_2\text{O} \text{ (kg)}}$$

$$= \frac{1/2}{0.8} = 0.625 \text{ m} \quad (\text{C}) \rightarrow \text{Correct}$$

$$\text{D) } M = \frac{\text{moles of solute}}{\text{vol. of solution (L)}}$$

$$= \frac{1/2}{0.8} = 0.625 \text{ M} \quad (\text{D}) \rightarrow \text{Correct}$$

Ans. $\rightarrow$ B, C, D (These are the correct options

in the answer key)

5. Ans (A,B,C,D)

$$(A) KE_{\max} = E_{\text{incident}} - \phi$$

$$\begin{aligned} \therefore (E_{\text{incident}})_{\min} &= KE_{\max} + \phi \\ &= (0.73 + 1.82) \text{ eV} \\ &= 2.55 \text{ eV} \quad (A) \rightarrow \text{correct} \end{aligned}$$

(B) $n = 4$ to $n = 2$ (H-atom)

$$\Delta E = 13.6 \times (1)^2 \left[\frac{1}{4} - \frac{1}{16} \right]$$

$$\Delta E = 13.6 \times \frac{3}{16} = 2.55 \text{ eV}$$

(Same as $E_{\text{incident}})_{\min}$

(B) $\rightarrow$ correct

(C) $n = 4$ (angular momentum)

$$(\ell_1) = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$$

$$n = 2 \quad \left(\ell_2 = \frac{2h}{2\pi} = \frac{h}{\pi} \right)$$

$$\therefore \text{change} \Rightarrow |\ell_2 - \ell_1| = \frac{h}{\lambda} \text{ (correct)}$$

$$(D) KE_{\text{collector}} = KE_{\text{emitter}} + (e \times 2V)$$

$\downarrow$

(due to potential diff.)

$$= 0.73 \text{ eV} + 2\text{eV}$$

$$= 2.73 \text{ eV} \text{ (Correct)}$$

Ans. : A, B, C, D

CHEMISTRY

SEC-3

3. Ans (6.00)



-



$$10^{-6} = \frac{y}{1 \times 1}$$

$$y = [A] = 10^{-6}$$

MATHEMATICS

SEC-1

1. Ans (B)

Let $\sin 2x = y$

$$(1 - \sin^2 2x)^2 + 2\sin^2 2x = 17(1 + \sin 2x)^4$$

$$(1 - y^2)^2 + 2y^2 = 17(1 + y)^4$$

$$1 + y^4 = 17(1 + y)^4$$

$$16 \left(y + \frac{1}{y} \right) + 68 \left(y + \frac{1}{y} \right) + 70 = 0$$

$$\text{put } y + \frac{1}{y} = z$$

$$\text{we get } 8z^2 + 34z + 35 = 0$$

$$z = -\frac{7}{4}, -\frac{5}{2}$$

$$\Rightarrow y + \frac{1}{y} = -\frac{5}{2} \quad y = -\frac{1}{2}, -2$$

$$\sin 2x = -\frac{1}{2}$$

4 solution $105^\circ, 165^\circ, 285^\circ, 345^\circ$

2. Ans (B)

$$C_1 : x^2 + y^2 - 8\sqrt{2}x - 6\sqrt{2}y + 25 = 0$$

$$\Rightarrow (x - 4\sqrt{2})^2 + (y - 3\sqrt{2})^2 = 25$$

Area is unaffected by change of origin

$$\Rightarrow x - 4\sqrt{2} = X$$

$$y - 3\sqrt{2} = Y$$

$$\Rightarrow C_1 : X^2 + Y^2 = 25;$$

$$P(4\sqrt{2}, 3\sqrt{2}); R(-4\sqrt{2}, -3\sqrt{2})$$

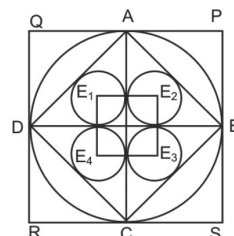
Now,

$\Rightarrow$ Observe that the points P and R lie on the circle $X^2 + Y^2 = 50$, which is the director circle of the given circle

$\Rightarrow$ Tangents from P & R are perpendicular to each other

Thus, PQRS is a square of side length 10

$\Rightarrow$ ABCD is also square of side lengths $5\sqrt{2}$



$E_1 E_2 E_3 E_4$ is also a square with side length

$$a = \frac{2\Delta}{s} = \frac{2 \left(\frac{1}{2} (5)^2 \right)}{\left(\frac{5+5+5\sqrt{2}}{2} \right)}$$

$$\text{Area } (E_1 E_2 E_3 E_4) = 50(3 - 2\sqrt{2})$$

3. Ans (D)

Here,

$$S = \sum_{r=1}^{16} \left(\left(\tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta + \frac{2\pi}{3}\right) + \tan\theta + \dots \right) \frac{(1 + \tan 4\theta \cdot \tan \theta)}{r} \right)$$

$$\therefore S = \sum_{r=1}^{16} \left((3 \tan 3\theta + 3 \tan 3\theta + \dots r \text{ times}) \cdot \frac{(1 + \tan 4\theta \cdot \tan \theta)}{r} \right)$$

$$= \sum_{r=1}^{16} 3 \cdot \tan 3\theta \cdot (1 + \tan 4\theta \cdot \tan \theta)$$

$$= \sum_{r=1}^{16} 3(\tan 4\theta - \tan \theta)$$

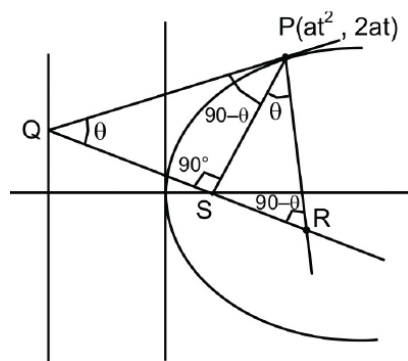
$$= 3 \sum_{r=1}^{16} (\tan 4^{r+1} - \tan 4^r)$$

$$= 3[(\tan 4^2 - \tan 4) + (\tan 4^3 - \tan 4^2) + \dots$$

$$(\tan 4^{17} - \tan 4^{16})]$$

$$\therefore S = 3(\tan 4^{17} - \tan 4)$$

4. Ans (A)



PQ subtend 90° at focus so $\angle RSP = 90^\circ$

(Also $\angle QPR = 90^\circ$)

From figure $\angle RQP = \angle SPR$, $\angle PRQ = \angle PRS$

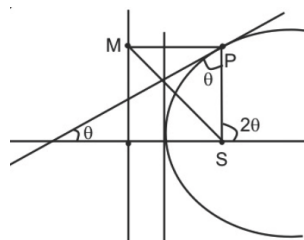
$$\Rightarrow \triangle PRQ \sim \triangle RSP \Rightarrow \frac{QR}{PR} = \frac{PQ}{PS} \Rightarrow$$

$$\frac{(PS)(QR)}{(PQ)(PR)} = 1$$

MATHEMATICS

SEC-2

1. Ans (A,B,C)



Slope of tangent is $\frac{1}{2} = \tan \theta$ (Let)

Slope of focal chord is

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2 \times \frac{1}{2}}{1 - \frac{1}{4}} = \frac{4}{3}$$

Equation of focal chord is

$$\frac{y-3}{x-4} = \frac{4}{3} \Rightarrow 4x - 3y - 7 = 0$$

Point of intersection of $4x - 3y - 7 = 0$ and x-

axis is $\left(\frac{7}{4}, 0\right) = \text{Focus}$

Image of focus on tangent will lie on directrix

which is $M\left(\frac{1}{4}, 3\right)$

Hence Directrix is $x = \frac{1}{4}$

2. Ans (A,B,C)

$$x > 0, x \neq 1 \text{ and } a \in (1, \sqrt{2}) \cup (\sqrt{2}, \infty)$$

$$\text{Also } 2a - x > 0 \Rightarrow 0 < x < 2a \Rightarrow x \in (0, 2a) - \{1\}$$

$$\text{now } \log_2(2a - x) + \log_2 x = \log_2(a^2 - 1)$$

$$\Rightarrow x(2a - x) = a^2 - 1$$

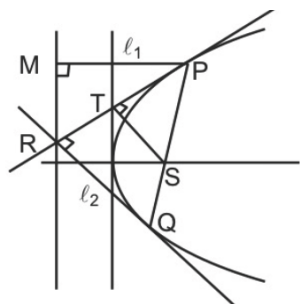
$$\Rightarrow x^2 - 2a + a^2 = 1 \Rightarrow (x - a)^2 = 1$$

$$\Rightarrow x = a \pm 1$$

$$x = a + 1 \text{ is true for all } a \in (1, \sqrt{2}) \cup (\sqrt{2}, \infty)$$

$$\text{and } x = a - 1 \text{ is true for } a \in (1, \infty) - \{\sqrt{2}, 2\}$$

3. Ans (A,C)



Let $PS = K_1x$ & $SQ = K_2x$

$$\text{Now } K_1 x = \ell_1 \Rightarrow x = \frac{\ell_1}{K_1}$$

$$SQ = \frac{K_2 \ell_1}{K_1} \text{ and } PS = \ell_1 \text{ (from figure PM = PS)}$$

PS, 2a, SQ are in HP.

$$\Rightarrow \text{semi L.R.} = \frac{2\text{PS.SQ}}{\text{PS} + \text{SQ}} = \frac{2\ell_1 \frac{K_2}{K_1} \ell_1}{\ell_1 + \frac{K_2}{K_1} \ell_1}$$

$$\Rightarrow \text{L. R.} = \frac{4K_2\ell_1}{K_1 + K_2}$$

$$\text{Now PQ} = \ell_1 + \frac{K_2 \ell_1}{K_1} = \frac{\ell_1}{K_1} (K_1 + K_2)$$

$$\Rightarrow \frac{\ell_1}{PO} = \frac{K_1}{K_1 + K_2}$$

$$\frac{\ell_1}{PQ} = \frac{\text{Distance of S from tangent}}{\ell_2} = \frac{ST}{\ell_2}$$

(because $\triangle PST$ and $\triangle PQR$ are similar)

From figure $PM = PS = \ell_1$

⇒ Distance of S from Tangent

$$= \text{ST} = \frac{\ell_1 \ell_2}{\ell_1 (\mathbf{K}_1 + \mathbf{K}_2)} \mathbf{K}_1 = \frac{\ell_2 \mathbf{K}_1}{\mathbf{K}_1 + \mathbf{K}_2}$$

4. Ans (A,B)

slope of tangent at any point is $\frac{a}{2y_1}$

Let $A(at_1^2, at_1)$ and $B(at_2^2, at_2)$

$t_1 t_2 = -\frac{1}{4}$ and centroid of

$$\Delta OAB = \left(\frac{a(s^2 + t^2)}{3}, \frac{a(s + t)}{3} \right)$$

so locus of centroid is $y^2 = \frac{a}{3} \left(x - \frac{a}{6} \right)$

so equation of parabola P_n

$$y^2 = \frac{a}{3^n} \left(x - \frac{a}{6} - \frac{a}{6 \cdot 3} - \frac{a}{6 \cdot 3^2} \cdots - \frac{a}{6 \cdot 3^{n-1}} \right)$$

$$y^2 = \frac{a}{3^n} \left(x - \frac{a}{6} \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} \right) \right)$$

$$y^2 = \frac{a}{3^n} \left(x - \frac{a}{6} \frac{\left(1 - \frac{1}{3^n}\right)}{\left(1 - \frac{1}{3}\right)} \right)$$

$$y^2 = \frac{a}{3^n} \left(x - \frac{a}{4} \left(1 - \frac{1}{3^n} \right) \right)$$

5. Ans (B,D)

$$\frac{a + ar + ar^2}{3} = \frac{7}{6} (a \cdot ar \cdot ar^2)^{1/3}$$

$$\Rightarrow 1 + r + r^2 = \frac{7}{2}r$$

$$\Rightarrow 2r^2 - 5r + 2 = 0 \Rightarrow r = 2, \frac{1}{2}$$

For $r = 2$ we have $\left\lceil \sqrt{\sqrt{2a}} \right\rceil = 1$

$$\Rightarrow 1 \leq 2a < 16 \Rightarrow \frac{1}{2} \leq a < 8$$

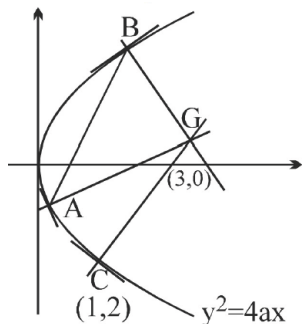
$$\Rightarrow a = 1, 2, 3, 4, 5, 6, 7$$

For $r = \frac{1}{2}$ we have $\frac{a}{2} \in [1, 16)$

$$\Rightarrow a \in [2, 32) \Rightarrow a = 2, 3, 4, \dots, 31$$

6. Ans (A,B,C)

Equation of AB



$$y = 2x - 1$$

solving it

$$y^2 = 4x$$

$$y^2 = 2(y + 1)$$

$$\Rightarrow y^2 - 2y - 2 = 0$$

but $y_1 + y_2 + y_3 = 0$

but $y_1 + y_2 = 2$

$$\therefore y_3 = -2$$

putting in $y^2 = 4x$

$$\mathbf{x}_3 = 1$$

hence coordinates of C are (1,-2)

$\therefore$ sum of the x and y coordinates of C are $-1 \Rightarrow$

(A) is correct.

obviously normal at C passes through the lower
end of the latus rectum

$\Rightarrow$ **(B)** is correct

again centroid of $\Delta ABC = \frac{x_1 + x_2 + x_3}{3}$

now solving

$$y = 2x - 1 \text{ with } y^2 = 4x$$

$$(2x - 1)^2 = 4x$$

$$4x^2 - 8x + 1 = 0 \Rightarrow x_1 + x_2 = 2; \text{ also } x_3 = 1$$

$$\therefore \text{centroid of the } \triangle ABC = \frac{2+1}{3} = (1, 0)$$

again equation of the normal at C

$$y + 2 = -\left(\frac{-2}{2}\right)(x - 1)$$

$$y + 2 = x - 1 \quad \Rightarrow \quad x - y - 3 = 0$$

hence gradient of chord at C is 1 $\Rightarrow$ **(D)** is incorrect

7. Ans (C,D)

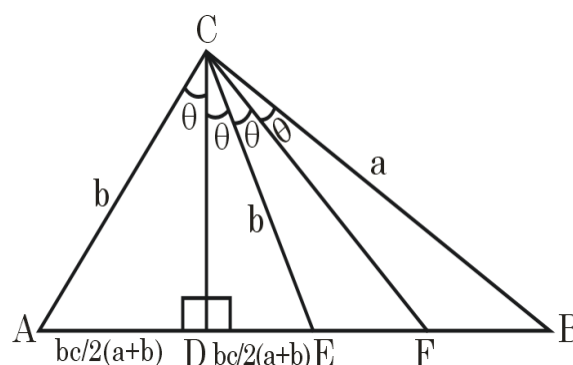
$$D_1 = \begin{vmatrix} -(x+1)^2 & (x^2-1) & \frac{1}{x-1} \\ x^2-1 & -(x-1)^2 & \frac{1}{x+1} \\ \frac{1}{x-1} & \frac{1}{x+1} & \frac{1}{(x-1)^2} \end{vmatrix}$$

where $x = 2^8$

$D_1 = D_2 = 4$ i.e., independent of 'x'.

8. Ans (D)

CE is angle bisector



$$\Rightarrow \frac{AE}{EB} = \frac{b}{a} \Rightarrow AE = \frac{bc}{a+b}$$

$$\Rightarrow EF = \frac{c}{2} - AE = \frac{c(a-b)}{2(a+b)}$$

$$\text{CF is angle bisector} \Rightarrow \frac{\text{CE}}{a} = \frac{\text{EF}}{\frac{c}{2}}$$

$$\Rightarrow \frac{b}{a} = \frac{(a-b)}{(a+b)} \Rightarrow ab + b^2 = a^2 - ab$$

$$\Rightarrow a^2 - 2ab - b^2 = 0$$

$$\Rightarrow \frac{a^2}{b^2} - \frac{2a}{b} - 1 = 0 \Rightarrow \frac{a}{b} = 1 + \sqrt{2}$$

$$\text{Also CE} = \frac{2ab}{a+b} \cos \frac{C}{2}$$

$$\Rightarrow b = \frac{2ab}{a+b} \cos \frac{C}{2}$$

$$\Rightarrow \frac{a+b}{a} = 2 \cos \frac{C}{2} \Rightarrow \frac{C}{2} = \frac{1}{\sqrt{2}} \Rightarrow C = 90^\circ$$

$$\Rightarrow A = \frac{\pi}{2} - \frac{\pi}{8} = \frac{3\pi}{8}$$

MATHEMATICS

SEC-3

1. Ans (0.50)

we can rewrite this equation as

$$\frac{x^2}{x^2-1} + \frac{x^2}{x^2-2} + \frac{x^2}{x^2-3} + \frac{x^2}{x^2-4} = \frac{1+(x^2-1)}{x^2-1} + \frac{2+(x^2-2)}{x^2-2} + \frac{3+(x^2-3)}{x^2-3} + \frac{4+(x^2-4)}{x^2-4}$$

$$= (2010x - 4) + 4 = 2010x$$

We divide by x ; this makes us lose the solution $x = 0$, but this does not affect the sum of solutions.

therefore, we have

$$x \{ (x^2-2)(x^2-3)(x^2-4) + (x^2-1)(x^2-3)(x^2-4) + (x^2-1)(x^2-2)(x^2-4) + (x^2-1)(x^2-2)(x^2-3) \}$$

$$= 2010(x^2-1)(x^2-2)(x^2-3)(x^2-4)$$

The solutions that we want are therefore the roots of the polynomial

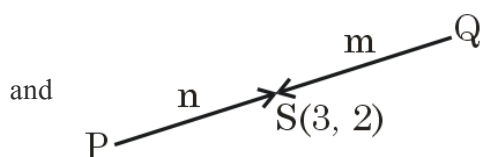
$$2010x^8 - 4x^7 + (\text{lower order terms}) = 0$$

BY Vieta's formulas, the sum of the roots of this polynomial equation is therefore $\frac{4}{2010}$

2. Ans (20.00)

P, Q and S are collinear with S in between the points P and Q. So PQ is a variable focal chord.

$$\begin{vmatrix} 3+2t & 0 & 1 \\ 4 & 2-1/t & 1 \\ 3 & 2 & 1 \end{vmatrix} = 0$$



$$\frac{4n + (3+2t)m}{m+n} = 3$$

$$\Rightarrow \frac{n}{m} = -2t > 0$$

A.M. $\geq$ H.M.

$$\Rightarrow \frac{PS + SQ}{2} \geq 2a \Rightarrow PQ \geq 4a$$

$$\Rightarrow PQ \geq 20$$

3. Ans (0.75)

$$\sqrt{3} \cos A \sin C = 2 - \sin A$$

Here $RHS > 0$

$LHS > 0 \Rightarrow \cos A$ is positive $\Rightarrow \angle A = \text{acute}$

$$\text{as } \sin C \leq 1 \Rightarrow \sqrt{3} \cos A \sin C \leq \sqrt{3} \cos A$$

$$2 - \sin A \leq \sqrt{3} \cos A$$

$$\sqrt{3} \cos A + \sin A \geq 2$$

$$\Rightarrow \sqrt{3} \cos A + \sin A \geq 2 \text{ it is only possible}$$

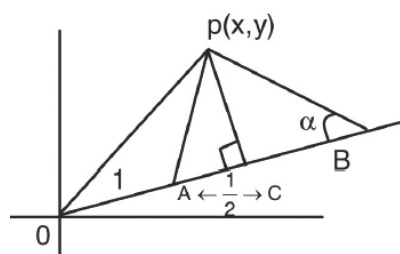
when

$$\sqrt{3} \cos A + \sin A = 2 \Rightarrow \angle A = \frac{\pi}{6}$$

$$\Rightarrow \sin C = 1 \Rightarrow \angle C = \frac{\pi}{2}, \angle B = \frac{\pi}{3}$$

$$\text{max area} = \frac{1}{2} \times 2 \times \sqrt{3} = \sqrt{3}$$

4. Ans (1.25)



C is the mid point of AB

$$AC = CB = \frac{1}{2} \Rightarrow OC = \frac{3}{2}$$

$$PC = \frac{\tan \alpha}{2}$$

$$\text{In } \triangle OPC \quad (OP)^2 = (OC)^2 + (PC)^2$$

$$x^2 + y^2 = \frac{9 + \tan^2 \alpha}{4}$$

5. Ans (0.80)

$$S = \sum_{n=1}^{24} \frac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}}$$

(Rationalize)

$$S = \sum_{n=1}^{24} \frac{(n+1)\sqrt{n} - n\sqrt{n+1}}{((n+1)\sqrt{n} - n\sqrt{n+1})((n+1)\sqrt{n} + n\sqrt{n+1})}$$

$$S = \sum_{n=1}^{24} \frac{(n+1)\sqrt{n} - n\sqrt{n+1}}{n(n+1)^2 - n^2(n+1)}$$

$$S = \sum_{n=1}^{24} \frac{(n+1)\sqrt{n} - n\sqrt{n+1}}{n(n+1)}$$

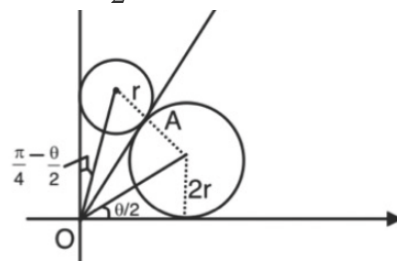
$$S = \sum_{n=1}^{24} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$$

$$= 1 - \frac{1}{\sqrt{25}} = 1 - \frac{1}{5} = \frac{4}{5}$$

6. Ans (5.50)

$$OA = r \cot\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \text{ and } OA = 2r \cot \frac{\theta}{2}$$

$$\text{Let, } \tan \frac{\theta}{2} = t$$



$$\Rightarrow \frac{1+t}{1-t} = \frac{2}{t}$$

$$\Rightarrow t^2 + 3t - 2 = 0$$

$$\Rightarrow t = \frac{-3 \pm \sqrt{17}}{2}$$

$$\therefore \tan \frac{\theta}{2} = \frac{\sqrt{17} - 3}{2}$$

$$\Rightarrow a + b + c = 17 + 3 + 2 = 22$$